

■ 3.7.10 Eigenvalues and Eigenvectors

<code>Eigenvalues[m]</code>	a list of the eigenvalues of m
<code>Eigenvectors[m]</code>	a list of the eigenvectors of m
<code>Eigensystem[m]</code>	a list of the form $\{\text{eigenvalues}, \text{eigenvectors}\}$
<code>Eigenvalues[N[m]]</code> , etc.	numerical eigenvalues
<code>Eigenvalues[N[m, k]]</code> , etc.	numerical eigenvalues, starting with k -digit precision

Eigenvalues and eigenvectors.

The eigenvalues of a matrix $\mathbf{m}$ are the values λ_i for which one can find vectors $\mathbf{x}_i$ such that $\mathbf{m} \cdot \mathbf{x}_i = \lambda_i \mathbf{x}_i$. The eigenvectors are the vectors $\mathbf{x}_i$.

Finding the eigenvalues of an $n \times n$ matrix in general involves solving an n^{th} degree polynomial equation. Since for $n \geq 5$ explicit algebraic solutions cannot in general be found, it is impossible to give explicit algebraic results for the eigenvalues and eigenvectors of all but the simplest or sparsest matrices.

Even for a matrix as simple as this, the explicit form of the eigenvalues is quite complicated.

```
In[1]:= Eigenvalues[ {{a, b}, {-b, 2a}} ]
Out[1]= { $\frac{3a + \text{Sqrt}[9a^2 - 4(2a^2 + b^2)]}{2}$ ,  

 $\frac{3a - \text{Sqrt}[9a^2 - 4(2a^2 + b^2)]}{2}$ }
```

If you give a matrix of approximate real numbers, *Mathematica* will find the approximate numerical eigenvalues and eigenvectors.

Here is a 2×2 numerical matrix.

```
In[2]:= m = {{2.3, 4.5}, {6.7, -1.2}}
Out[2]= {{2.3, 4.5}, {6.7, -1.2}}
```

The matrix has two eigenvalues, in this case both real.

```
In[3]:= Eigenvalues[m]
Out[3]= {6.31303, -5.21303}
```

Here are the two eigenvectors of $\mathbf{m}$.

```
In[4]:= Eigenvectors[m]
Out[4]= {{0.746335, 0.665557}, {-0.523116, 0.873374}}
```

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`Eigensystem` computes the eigenvalues and eigenvectors at the same time. The assignment sets `vals` to the list of eigenvalues, and `vecs` to the list of eigenvectors.

```
In[5]:= {vals, vecs} = Eigensystem[m]
Out[5]= {{6.31303, -5.21303},
         {{0.746335, 0.66557}, {-0.523116, 0.873374}}}
```

This verifies that the first eigenvalue and eigenvector satisfy the appropriate condition.

```
In[6]:= m . vecs[[1]] == vals[[1]] vecs[[1]]
Out[6]= True
```

This finds the eigenvalues of a random 4×4 matrix. For non-symmetric matrices, the eigenvalues can have imaginary parts.

```
In[7]:= Eigenvalues[ Table[Random[ ],{i,4},{j,4}] ]
Out[7]= {1.76797, -0.235343 + 0.433364 I,
         -0.235343 - 0.433364 I, 0.432073}
```

The function `Eigenvalues` always gives you a list of n eigenvalues for an $n \times n$ matrix. The eigenvalues correspond to the roots of the characteristic polynomial for the matrix, and may not necessarily be distinct. `Eigenvectors`, on the other hand, gives a list of eigenvectors which are guaranteed to be independent. The number of such eigenvectors may be less than n .

Here is a 3×3 matrix.

```
In[8]:= mz = {{0, 1, 0}, {0, 0, 1}, {0, 0, 0}}
Out[8]= {{0, 1, 0}, {0, 0, 1}, {0, 0, 0}}
```

The matrix has three eigenvalues, all equal to zero.

```
In[9]:= Eigenvalues[mz]
Out[9]= {0, 0, 0}
```

There is, however, only one independent eigenvector for the matrix.

```
In[10]:= Eigenvectors[mz]
Out[10]= {{1, 0, 0}, {0, 0, 0}, {0, 0, 0}}
```